

- The slope of the S-line is MPS, which is (1 – slope of C-line).

Investment Multiplier

It measures the change in income due to a change in investment in the economy. It is denoted by 'K'.

$$K = \frac{\Delta Y}{\Delta I}$$

Also, $K = \frac{1}{MPS}$ or $K = \frac{1}{1-MPC}$

- Maximum value of K is infinity (when $MPS = 0$).
- Minimum value of K is 1 (when $MPS = 1$).

Relationship between MPC, MPS and K

MPC	K	MPS	K
0.2	1.25	0.8	1.25
0.4	1.67	0.6	1.67
0.5	2	0.5	2
0.8	5	0.2	5

$$MPC + MPS = 1$$

$$K = \frac{1}{MPS} \text{ or } K = \frac{1}{1-MPC}$$

Conclusion on Multiplier:

- Inverse relationship between multiplier and MPS: Higher MPS leads to lower K.
- Direct relationship between multiplier and MPC: Higher MPC leads to higher K.

Working of Investment Multiplier (Numerical Example)

If $\Delta I = 100$ and $MPC = 0.8$, then $K = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$.

$$\text{So, } \Delta Y = K \times \Delta I = 5 \times 100 = 500.$$

The initial increase in investment of 100 crores results in an additional income of 500 crores.

Key Terms

- **Ex-ante saving:** Planned saving in an economy in a year.
- **Ex-ante investment:** Planned investment in an economy in a year.
- **Ex-post investment:** Actual amount of investment in an economy during a year.
- **Ex-post saving:** Actual savings in an economy in a year.